Frobenius-equivariant pair extension and robust repair of eight-arcs

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July 2026

Abstract

Let ϕ be the quadratic Frobenius involution of $\mathrm{PG}(2, s^2)$. We study extensions of ϕ-invariant arcs by fresh nonfixed conjugate pairs, or equivalently Frobenius-compatible two-column extensions of the associated dimension-three MDS codes. Every pair lies on a unique fixed mate line. Counting pairs carrier by carrier gives a uniform lower bound, and an exact correction records secant orbits that disappear at fixed centers and visible orbits that collide on one candidate.

Every invariant eight-arc over every prime-power base order $s \geq 5$ admits such an extension. Over $\mathbb{F}_{25}$ there are at least four legal pairs; the exceptional two-fixed-point profile has exact minimum 32. For $s \geq 7$ there are at least 319 legal pairs. These bounds yield alternate repairs after deletion of a selected orbit, and a parameterized criterion gives the corresponding result for invariant $(k+2)$-arcs. The structural reductions have human-scale Lean support. A separate Mathlib-only certificate checks the normalized exceptional census; the projective normalizations and semantic transport are manuscript arguments.

1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. Ordinary completeness asks whether one further point can be adjoined. If the arc is preserved by a field automorphism, the natural equivariant continuation unit is instead a complete Galois orbit. Over a quadratic extension this means either a fixed point or a nonfixed conjugate pair.

Let $F = \mathbb{F}_s$, let $K = \mathbb{F}_{s^2}$, and let

$$\phi\colon [x : y : z] \longmapsto [x^s : y^s : z^s]$$

act on $\mathrm{PG}(2, K)$. The fixed points and fixed lines form the embedded Baer subplane $\mathrm{PG}(2, F)$. This paper studies the following question: when must a ϕ-invariant arc admit an extension by a fresh pair $\{P, \phi(P)\}$?

Under the arc–code correspondence, a k-arc gives the projective columns of a $[k, 3, k-2]_{s^2}$ MDS code. Adjoining $\{P, \phi(P)\}$ is therefore a Frobenius-compatible two-column MDS extension, and alternate-orbit repair is replacement of an erased conjugate column pair. Here "repair" changes the generator-column configuration; it is not erasure decoding inside a fixed code.

Theorem 1.1 (Robust extension of invariant eight-arcs). *Let $s \geq 5$ be a prime power. Every ϕ-invariant eight-arc in $\mathrm{PG}(2, s^2)$ admits a fresh nonfixed conjugate-pair extension. More precisely:*

(i) over $\mathbb{F}_{25}$ there are at least four such pairs, so deleting any selected nonfixed orbit from an invariant ten-arc leaves at least three alternate repairs;

(ii) for every $s \geq 7$ there are at least 319 *legal pairs, so deletion of any selected nonfixed orbit from an invariant ten-arc leaves at least* 318 *alternate repairs.*

The proof has one organizing mechanism. A nonfixed pair lies on a unique fixed mate line, and a legal pair must lie on a fixed line disjoint from the old arc. Each noninvariant old-secant orbit forbids at most one candidate on such a carrier. Counting empty carriers gives the uniform criterion. The exact correction records the two ways in which the first-order count overestimates the obstruction: the secant orbit can be invisible because its fixed center lies on the carrier, or several visible orbits can charge the same pair.

The argument first proves the carrier count and its exact correction. For $s \geq 7$ those structural bounds imply Theorem 1.1(ii). At $s = 5$ one profile lies beyond the first-order range; two projective normalizations reduce it to a finite residual problem. Proposition 5.3 gives the exact minimum 32 in that profile and classifies its normalized equality cases. Combining it with the four structural profiles proves Theorem 1.1(i).

The quantitative criterion and its exact correction are the structural part of the paper. Proposition 5.3 is the only exhaustive finite step. Its lower bound, attainment, and equality-orbit exhaustion are kernel checked in the normalized coordinate model; the manuscript carries those facts through the two projective normalizations. Section 7 states the certificate and trust boundary. The parameterized exchange theorem identifies the general phase $\lfloor (k-1)^2/4 \rfloor + r + 1 \leq s(s-1)/2$. The saturation corollary and the Clebsch specialization are secondary consequences of the same carrier count. The saturation scale is the classical Lunelli–Sce $\sqrt{2q}$ scale [13, 5]; no new constant or sharpness claim is made.

The classical inputs are point and line counts in a projective plane, the fixed Baer subplane, Hilbert–90 normalization, and secant counting; see Hirschfeld [10] and Martin [15].

Prior art and claim boundary

Baker and Wantz [4] use Frobenius-invariant arcs in the Hughes plane and, in a maximality argument, consider a point together with its Frobenius image. This is genuine precedence for the conjugate-point maneuver. Their result does not count legal conjugate pairs on empty Baer lines. Jungnickel's arcs in Baer-semiplane complements [11] and Ueberberg's geometries of Frobenius collineations [19] are also adjacent but concern different incidence or collineation problems.

Dye gives exact completion geometry for the special family of Clebsch hexagons and a five-valent adjacency graph defined by sharing an associated triangle [9, Proposition 1 and Theorems 7–8, pp. 282–284]. Blokhuis, Seress, and Wilbrink classify extremal complete exterior sets relative to a conic [6, pp. 143–146]. Neither work treats arbitrary plane arcs invariant under the quadratic field-Frobenius involution, counts fresh legal conjugate-pair extensions, or requires a different replacement after every selected-orbit deletion.

Ordinary arc and MDS lengthening results [1, 3, 2] impose no quadratic Galois-pair constraint. Likewise, the classification of Galois-invariant subsets of $\mathbf{P}^1$ [12] does not treat arbitrary plane arcs or their pair extensions. Bounded zbMATH Open, OpenAlex, Crossref, and source-level searches found no exact precursor for the quantitative criterion below for arbitrary quadratic- field-Frobenius-invariant arcs, or for the uniform $\mathrm{PG}(2, 25)$ theorem. This is negative search evidence, not a claim of historical priority.

2 Quadratic Frobenius and mate lines

Let $\mathcal{C} \subseteq \mathrm{PG}(2, K)$ be a ϕ-invariant k-arc. Write

$$f = |\mathcal{C} \cap \mathrm{PG}(2, F)|, \qquad k = f + 2e,$$

so that $\mathcal{C}$ is the disjoint union of f fixed points and e nonfixed conjugate pairs.

For every nonfixed point P, the line $P\phi(P)$ is fixed by ϕ; we call it the *mate line* of the pair $\{P, \phi(P)\}$. Conversely, every nonfixed conjugate pair has a unique mate line. A fixed line contains s^2+1 points, of which $s+1$ are fixed, and therefore contains exactly

$$N = \frac{s^2 - s}{2}$$

nonfixed conjugate pairs.

Definition 2.1. A *legal conjugate-pair extension* of $\mathcal{C}$ is an unordered pair $Q = \{P, \phi(P)\}$ such that $P \neq \phi(P)$, $Q \cap \mathcal{C} = \varnothing$, and $\mathcal{C} \cup Q$ is an arc. Let $N_{\text{pair}}(\mathcal{C})$ be the number of such pairs.

Definition 2.2 (Alternate-orbit repair)**.** Let A be an invariant arc, let $O \subseteq A$ be a selected nonfixed conjugate orbit, and put $D = A \setminus O$. An *alternate-orbit repair* of this deletion is a legal conjugate-pair extension Q of D with $Q \neq O$. The erased orbit O is itself legal for D, so a lower bound $N_{\text{pair}}(D) \geq r+1$ supplies at least r alternate repairs.

Lemma 2.3 (Pair-extension test)**.** *Let $P \neq \phi(P)$ and suppose neither point belongs to $\mathcal{C}$. Then $\mathcal{C} \cup \{P, \phi(P)\}$ fails to be an arc if and only if at least one of the following holds:*

(a) P lies on a secant of $\mathcal{C}$;

(b) $\phi(P)$ lies on a secant of $\mathcal{C}$;

(c) the mate line $P\phi(P)$ contains a point of $\mathcal{C}$.

Proof. Every collinear triple in the enlarged set either contains one new point and two old points, giving (a) or (b), or contains both new points and one old point, giving (c). The converse is immediate. □

The useful carriers are therefore the fixed lines disjoint from $\mathcal{C}$. Let $\mathcal{E}$ be this set of empty fixed lines.

Lemma 2.4 (Empty fixed lines)**.** *The number of empty fixed lines is*

$$E = |\mathcal{E}| = s^2 + s + 1 - \left(f(s+1) - \binom{f}{2} + e \right). \tag{1}$$

Proof. The f fixed selected points occupy $f(s+1) - \binom{f}{2}$ fixed lines. Inclusion–exclusion is exact because no three selected fixed points are collinear. Each selected conjugate pair occupies its mate line. These e lines are distinct and contain no fixed selected point, since $\mathcal{C}$ is an arc. Subtract from the s^2+s+1 fixed lines. □

Among the $\binom{k}{2}$ old secants, exactly

$$I = \binom{f}{2} + e$$

are fixed: those joining two fixed selected points and the e mate lines. The remaining secants occur in two-element Frobenius orbits. Their number is

$$M = \frac{\binom{k}{2} - I}{2} = fe + e(e-1). \tag{2}$$

3 The quantitative pair-extension criterion

The mate-line decomposition now turns the extension problem into one count: empty fixed lines are the carriers, and nonfixed secant orbits are the only first-order obstructions on them.

Theorem 3.1 (Quadratic pair-extension criterion)**.** *Let $\mathcal{C}$ be a ϕ-invariant k-arc in* $\mathrm{PG}(2,s^2)$ *with profile $k = f + 2e$. With E, N, and M as in* (1) *and* (2)*,*

$$N_{\text{pair}}(\mathcal{C}) \geq E\ (N-M)_+ = E\left(\frac{s^2-s}{2} - fe - e(e-1)\right)_+ . \tag{3}$$

In particular, if $E > 0$ and $M < N$, then $\mathcal{C}$ admits a legal conjugate-pair extension.

Proof. Fix $\ell \in \mathcal{E}$. It carries exactly N candidate conjugate pairs. A fixed old secant meets ℓ in a fixed point and destroys no nonfixed candidate. Consider a nonfixed old secant orbit $\{m, \phi(m)\}$. Unless $m \cap \ell$ is fixed, the two intersections $m \cap \ell$ and $\phi(m) \cap \ell$ form one conjugate candidate pair; thus the orbit destroys at most one candidate. There are M such secant orbits, so at least $(N-M)_+$ candidates survive on ℓ.

A survivor lies on no old secant, and its empty mate line contains no point of $\mathcal{C}$. Lemma 2.3 makes it legal. Candidate sets on distinct fixed lines are disjoint because a nonfixed pair has a unique mate line. Summing over the E carriers proves (3). ☐

Corollary 3.2 (The Clebsch hexagon over the quadratic extension)**.** *Let $H \subseteq$* $\mathrm{PG}(2,11)$ *be a Clebsch hexagon, viewed as a Frobenius-invariant six-arc in* $\mathrm{PG}(2,11^2)$ *[17]. Then*

$$N_{\text{pair}}(H) = 76 \cdot 55 = 4180.$$

Consequently every invariant eight-arc $H \cup Q$ obtained by adjoining one such pair has exactly 4179 *alternate conjugate-pair repairs after Q is deleted. In coding language, the scalar extension of the Clebsch* $[6,3,4]_{11}$ *code has* 4180 *Frobenius-paired two-column extensions to* $[8,5,4]_{121}$ *MDS codes.*

Proof. Here $(s,f,e) = (11,6,0)$. There are

$$E = 11^2 + 11 + 1 - \left(6 \cdot 12 - \binom{6}{2}\right) = 76$$

fixed lines disjoint from H, and each contains $N = (11^2-11)/2 = 55$ nonfixed conjugate pairs. Every old secant is fixed, so $M = 0$ and every one of these pairs is legal. Conversely, the mate line of any legal nonfixed pair is a fixed line disjoint from H, which proves equality rather than only the lower bound. Excluding the selected pair Q leaves $4180 - 1$ alternatives. ☐

Remark 3.3. The count in Corollary 3.2 uses only that H is a six-arc rational over $\mathbb{F}_{11}$; it is not another characterization of the Clebsch class. Its role is to show how the exceptional code studied in [17] enters the equivariant extension and repair theory developed here.

The preceding proof deliberately uses only the uniform upper bound M on the number of forbidden candidates per carrier. Retaining the actual forbidden support gives both a sharper bound and an exact identity.

For $\ell \in \mathcal{E}$, let $\mathcal{Q}_\ell$ be its N candidate pairs and let $\mathcal{F}_\ell \subseteq \mathcal{Q}_\ell$ be the pairs hit by at least one nonfixed old secant orbit. Then

$$N_{\text{pair}}(\mathcal{C}) = \sum_{\ell \in \mathcal{E}} \left(|\mathcal{Q}_\ell| - |\mathcal{F}_\ell|\right). \tag{4}$$

The same formula applies when carrier capacities vary. In particular, if $|\mathcal{F}_\ell| \leq U_\ell$, then

$$N_{\text{pair}}(\mathcal{C}) \geq \sum_{\ell \in \mathcal{E}} \left(|\mathcal{Q}_\ell| - U_\ell\right)_+ .$$

4 Invisible centers and collision correction

The preceding theorem subtracts every nonfixed secant orbit once on every empty carrier. Exactness requires recording when an orbit contributes nothing and when several orbits forbid the same candidate.

Fix $\ell \in \mathcal{E}$. Every nonfixed secant orbit has a fixed center $m \cap \phi(m)$. Let B_ℓ be the number of such orbits whose center lies on ℓ. Those orbits meet ℓ only in a fixed point and are invisible to its nonfixed candidates.

For $Q \in \mathcal{Q}_\ell$, let $\mu_\ell(Q)$ be the number of visible secant orbits that charge Q, and put

$$A_\ell = \sum_{Q \in \mathcal{Q}_\ell} \mu_\ell(Q), \qquad R_\ell = \sum_{Q \in \mathcal{Q}_\ell} (\mu_\ell(Q) - 1)_+ .$$

Here R_ℓ is the redundancy caused by several visible obstruction orbits hitting the same candidate. Figure 1 separates this collision from an invisible orbit whose fixed center lies on the carrier.

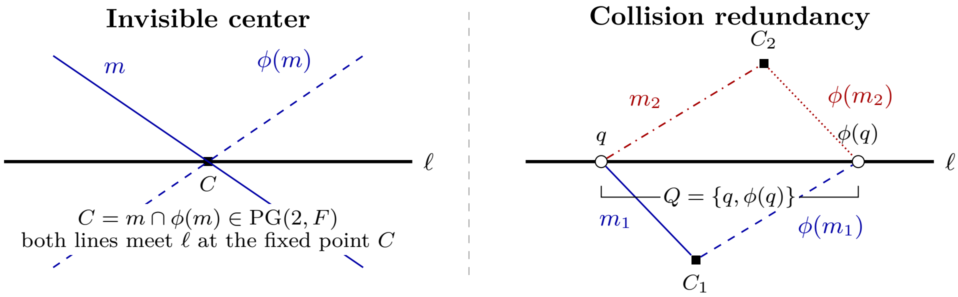

Figure 1: The two corrections to the first-order obstruction count on an empty fixed carrier ℓ. Left: when the fixed center $C = m \cap \phi(m)$ lies on ℓ, the conjugate secants meet the carrier only at a fixed point and charge no candidate pair. Right: two distinct visible secant orbits have different fixed centers but charge the same candidate $Q = \{q, \phi(q)\}$; only one forbidden pair remains in the support. Solid/dashed and dash-dot/dotted lines distinguish the conjugate secants in the two visible orbits.

Proposition 4.1 (Exact linewise correction)**.** *For every empty fixed line ℓ,*

$$A_\ell = M - B_\ell, \tag{5}$$
$$|\mathcal{F}_\ell| = A_\ell - R_\ell, \tag{6}$$
$$|\mathcal{Q}_\ell \setminus \mathcal{F}_\ell| + M = N + B_\ell + R_\ell. \tag{7}$$

Consequently the first-order identity $|\mathcal{Q}_\ell \setminus \mathcal{F}_\ell| + M = N$ holds exactly when every secant orbit is visible on ℓ and the visible charge map is injective.

Proof. Exactly B_ℓ of the M secant orbits are invisible, proving (5). The support of the multiplicity function μ_ℓ is $\mathcal{F}_\ell$, and for every positive integer a one has $1 = a - (a-1)$. Summing this equality over the support gives (6). Substitution into $|\mathcal{Q}_\ell \setminus \mathcal{F}_\ell| = N - |\mathcal{F}_\ell|$ gives (7). Equality in the first-order identity is equivalent to $B_\ell = R_\ell = 0$. □

Let

$$L = N_{\text{pair}}(\mathcal{C}), \qquad B = \sum_{\ell \in \mathcal{E}} B_\ell, \qquad R = \sum_{\ell \in \mathcal{E}} R_\ell.$$

Corollary 4.2 (Aggregate equality and excess)**.** *The exact aggregate balance is*

$$L + EM = EN + B + R. \tag{8}$$

Thus $L + EM = EN$ if and only if every secant-orbit center avoids every empty fixed carrier and every local visible charge is injective. More generally, for each integer $t \geq 0$,

$$L + EM = EN + t \quad \iff \quad B + R = t.$$

This is an algebraic equality and excess classification. It does not classify the invariant arcs for which L is small, and no such structural inverse theorem is claimed.

5 Eight-arcs over $\mathbb{F}_{25}$

We now take $s = 5$, so each empty fixed line carries $N = 10$ candidate pairs. The profile of an invariant eight-arc is

$$(f, e) \in \{(0,4), (2,3), (4,2), (6,1), (8,0)\}.$$

The four profiles with $f \neq 2$ are closed by incidence and moment arguments. Only $(f, e) = (2, 3)$ escapes the structural bounds. Table 1 separates these routes; the exceptional row records the exact kernel-checked minimum and equality classification after normalization.

f	e	E	M	proved $L \geq$	method
0	4	27	12	5	secant-index moments
2	3	17	12	32	exact minimum and five-orbit classification
4	2	11	10	4	fixed-center incidence
6	1	9	6	36	uniform criterion
8	0	11	0	110	uniform criterion

Table 1: The five profiles of invariant eight-arcs in $\mathrm{PG}(2, 25)$. Here $L = N_{\mathrm{pair}}(\mathcal{C})$.

The four structural profiles

The following simple center estimate is useful. A secant orbit joining two distinct selected conjugate pairs has a fixed center that lies on neither participating mate line. At most f fixed-point star lines and $e - 2$ other mate lines through this center are occupied. Since a fixed point lies on $s + 1$ fixed lines, each such cross-pair secant orbit is invisible on at least

$$s + 3 - f - e \tag{9}$$

empty carriers.

Proposition 5.1 (The four nonexceptional profiles)**.** *Let $\mathcal{C}$ be an invariant eight-arc in* $\mathrm{PG}(2, 25)$.

(a) If $f = 8$ or $f = 6$, then $L \geq 110$ or $L \geq 36$, respectively.

(b) If $f = 4$, then $L \geq 4$.

(c) If $f = 0$, then $L \geq 5$.

Proof. For $(f, e) = (8, 0)$ one has $(E, M) = (11, 0)$, and for $(6, 1)$ one has $(E, M) = (9, 6)$. Theorem 3.1 gives 110 and 36.

For $(f, e) = (4, 2)$ one has $(E, M, N) = (11, 10, 10)$. There are two cross-pair secant orbits, each invisible on at least two empty carriers by (9). Hence $B \geq 4$, and (8) gives $L = B + R \geq 4$.

It remains to treat $(f, e) = (0, 4)$. Here

$$(E, M, N) = (27, 12, 10).$$

All twelve nonfixed secant orbits are cross-pair orbits, and (9) gives $B \geq 48$.

We recall the second secant-index moment for an eight-arc:

$$\sum_{x \notin \mathcal{C}} \binom{r_x}{2} = 3\binom{8}{4} = 210, \tag{10}$$

where r_x is the number of secants of $\mathcal{C}$ through x. At fixed external points, the first secant-index sum is 48: the four fixed mate secants contain six fixed points each, contributing 24 incidences, and the other 24 secants each contain their unique fixed orbit center. Since $r_x \leq 4$ and $2\binom{n}{2} \leq 3n$ for $0 \leq n \leq 4$, their contribution to (10) is at most 72.

The four occupied fixed carriers are precisely the selected mate lines. For an external nonfixed point x on one of them, write $r_x = 1 + u_x$, where the first secant is the mate line. Every nonfixed secant meets at most the two selected mate lines not containing its endpoints, so $\sum u_x \leq 48$. Since $u_x \leq 3$ and $\binom{1+u}{2} \leq 2u$, the occupied-carrier contribution is at most 96. Thus the two endpoints of candidates on empty carriers contribute at least $210 - 72 - 96 = 42$ to (10). Conjugate endpoints contribute equally, and therefore

$$T := \sum_{\ell \in \mathcal{E}} \sum_{Q \in \mathcal{Q}_\ell} \binom{\mu_\ell(Q)}{2} \geq 21.$$

Because $\mu_\ell(Q) \leq 4$ and $\binom{n}{2} \leq 2(n-1)$ for $1 \leq n \leq 4$, we have $T \leq 2R$, whence $R \geq 11$. Finally, (8) gives

$$L = 27(10 - 12) + B + R \geq -54 + 48 + 11 = 5.$$

□

Corollary 5.2 (Nonexceptional order-five alternate repair). *Let A be an invariant ten-arc in* $\mathrm{PG}(2, 25)$*, let $O \subseteq A$ be a selected nonfixed conjugate orbit, and put $D = A \setminus O$. If the fixed point count of D is $f = 0, 4, 6, 8$, then this deletion has at least* $4, 3, 35, 109$ *alternate-orbit repairs, respectively.*

Proof. The set D is an invariant eight-arc and O is one of its legal conjugate-pair extensions. Proposition 5.1 gives the four total legal-pair bounds $5, 4, 36, 110$. Removing the one possibility O gives the asserted alternate counts. □

The remaining profile $(f, e) = (2, 3)$ has $(E, M, N) = (17, 12, 10)$. The center estimate gives $B \geq 18$, but the aggregate identity would still require a further collision estimate. We close this profile by a finite reduction.

The exceptional two-fixed-point profile

Proposition 5.3 (Exact exceptional minimum). *Every invariant eight-arc in* $\mathrm{PG}(2, 25)$ *with exactly two fixed selected points has at least* 32 *fresh legal conjugate-pair extensions. This* *bound is attained. After the two projective normalizations used below, the configurations attaining equality form exactly five residual-group orbits, of sizes*

$$200, \qquad 400, \qquad 400, \qquad 200, \qquad 400.$$

Their disjoint union has size 1600*. Every normalized configuration outside this union has at least* 33 *legal pairs.*

Computer-assisted proof, kernel checked. Represent $\mathbb{F}_{25}$ as $\mathbb{F}_5[\omega]/(\omega^2 - 2)$. A base-field projectivity sends the two fixed selected points to a prescribed ordered pair. Their stabilizer then sends the first admissible selected conjugate pair to

$$[1 : \omega : \omega], \qquad [1 : -\omega : -\omega].$$

Both transformations commute with Frobenius and preserve the arc and pair-extension predicates.

Here $(E, M, N) = (17, 12, 10)$, so (8) reads $L = B + R - 34$ and the target is the coupled inequality $B + R \geq 66$. The finite layer below is load bearing for this coupling; separate bounds on invisible centers and collision moments do not prove it.

After these reductions, the finite slice has 46,056 rows and 1,189 residual class representatives. Kernel-checked composition certificates give a lower bound of 32 on every representative. Five representatives attain 32. The residual group has order 400; orbit–stabilizer certificates give the five displayed orbit sizes, prove their pairwise disjointness, and show that their union has size 1600.

Strict certificates place every nonminimizing residual class above 32, and an exhaustive dispatcher places every normalized equality row in the five-orbit union. The two projective normalizations and invariance of the legal-pair count carry the lower bound and equality exhaustion back to the semantic arc. Conversely, invariance of the legal-pair count under the residual action transports the five checked equalities across their orbits. Thus 32 is the exact semantic minimum, with the stated complete equality classification up to normalization. Section 7 records the trust boundary and the generated-certificate layers. □

Theorem 5.4 (Uniform order-five multiplicity)**.** *Every Frobenius-invariant eight-arc in* $\mathrm{PG}(2, 25)$ *has at least four distinct fresh nonfixed conjugate-pair extensions.*

Proof. The fixed-point count is even and at most eight. Proposition 5.1 gives respective lower bounds $5, 4, 36, 110$ for $f = 0, 4, 6, 8$, and Proposition 5.3 gives 32 for $f = 2$. □

Corollary 5.5 (Uniform order-five alternate repair)**.** *Let A be an invariant ten-arc in* $\mathrm{PG}(2, 25)$ *and let $O \subseteq A$ be any selected nonfixed conjugate orbit. After deleting O, at least one different legal conjugate orbit repairs the arc; in fact at least three do.*

Proof. The remainder $D = A \setminus O$ is an invariant eight-arc, and O is one of its legal conjugate-pair extensions. Theorem 5.4 supplies at least four legal pairs for D, so at least three are different from O. □

The complete-arc classifications in [14, 7] show that the smallest complete arcs in $\mathrm{PG}(2, 25)$ have size 12. Thus every eight-arc has an ordinary one-point extension. That fact does not imply Theorem 5.4: ordinary extendability neither makes the legal-point set Frobenius-stable nor ensures that a legal point and its conjugate are jointly legal.

Remark 5.6 (Scope of the equality classification). Proposition 5.3 classifies equality up to normalization: it says that some base-field collineation sends an equality case into the certified 1600-element union. It neither asserts uniqueness of that collineation nor enumerates the semantic arcs attaining 32. It makes no equality claim for the other fixed-point profiles, other arc sizes, or other fields.

6 General robust repair

The profile envelope and parameterized exchange theorem are the general repair statements of the paper. We first record a secondary saturation consequence of the carrier criterion, then derive those two results.

A saturation consequence

Call an invariant arc *pair-saturated* if it has no legal conjugate-pair extension. This is weaker than ordinary completeness.

Corollary 6.1 (Orbit-saturation bound)**.** *If a ϕ-invariant k-arc in* $\mathrm{PG}(2, s^2)$ *is pair-saturated, then*

$$k \geq 1 + \left\lceil \sqrt{2s(s-1)} \right\rceil . \tag{11}$$

In particular, for every prime power $s \geq 7$, every invariant eight-arc in $\mathrm{PG}(2, s^2)$ *admits a conjugate-pair extension.*

Proof. If every fixed line is occupied, Lemma 2.4 gives $s^2 + s + 1 = f(s+1) - \binom{f}{2} + e$. Substituting $2e = k - f$ and completing the square gives the occupation identity

$$k = s^2 + 1 + (f - s - 1)^2,$$

which is stronger than (11). Otherwise choose an empty fixed line. Pair saturation forces all $N = s(s-1)/2$ of its candidates to be forbidden, so $N \leq M$. From $k = f + 2e$ and (2),

$$M = e((k-1) - e), \qquad 4M \leq (k-1)^2.$$

Consequently $2s(s-1) \leq (k-1)^2$, which gives (11).

For $k = 8$, one has $M \leq 12 < N$ when $s \geq 7$. Full occupation is impossible because $8 < s^2 + 1$, so Theorem 3.1 applies. □

Profile envelope and robust exchange

Theorem 6.2 (Five-profile lower envelope)**.** *Let $s \geq 7$ and let D be a ϕ-invariant eight-arc with profile $8 = f + 2e$. Put $N = s(s-1)/2$. Then $N_{\mathrm{pair}}(D) \geq L_f(s)$, where*

(f, e)	$E_f(s)$	M_f	$L_f(s) = E_f(s)(N - M_f)$
$(0, 4)$	$s^2 + s - 3$	12	$(s^2 + s - 3)(N - 12)$
$(2, 3)$	$s^2 - s - 3$	12	$(s^2 - s - 3)(N - 12)$
$(4, 2)$	$s^2 - 3s + 1$	10	$(s^2 - 3s + 1)(N - 10)$
$(6, 1)$	$s^2 - 5s + 9$	6	$(s^2 - 5s + 9)(N - 6)$
$(8, 0)$	$s^2 - 7s + 21$	0	$(s^2 - 7s + 21)N.$

For the certified first-order envelope $L(s) = \min_f L_f(s)$,

$$L(7) = 319, \qquad L(8) = 726, \qquad L(9) = 1350,$$

and $L(s) = L_8(s)$ for every $s \geq 10$. The minimizing profiles are therefore $f = 4$ at $s = 7$, $f = 6$ at $s = 8, 9$, and $f = 8$ for $s \geq 10$. In particular, $N_{\mathrm{pair}}(D) \geq 319$ for every $s \geq 7$.

Proof. Substituting the five solutions of $8 = f + 2e$ into (1) and (2) gives the displayed values of E_f, M_f, and hence $L_f = E_f(N - M_f)$ by Theorem 3.1. At $s = 7, 8, 9$ the five value vectors are respectively

$$(477, 351, 319, 345, 441), \quad (1104, 848, 738, 726, 812), \quad (2088, 1656, 1430, 1350, 1404).$$

For $s = t + 10$, twice each difference $L_f - L_8$, for $f = 0, 2, 4, 6$, has nonnegative coefficients as a polynomial in $t \geq 0$. This proves the infinite branch. Independently, each $L_f(s)$ is nondecreasing for $s \geq 7$, so the value 319 at the first boundary is a uniform lower bound. These polynomial identities and inequalities are kernel-checked in the supplementary Lean source. □

Corollary 6.3 (Quantitative alternate-orbit repair). *Let $s \geq 7$ be a prime power. If A is a ϕ-invariant ten-arc in* $\mathrm{PG}(2, s^2)$ *and $O \subseteq A$ is any selected nonfixed conjugate orbit, then deleting O leaves at least* 318 *alternate-orbit repairs.*

Proof. Put $D = A \setminus O$. Then D is an invariant eight-arc, and O is one of its legal conjugate-pair extensions. Theorem 6.2 gives $N_{\text{pair}}(D) \geq 319$. Excluding O leaves at least 318 different legal pairs. □

Theorem 6.4 (Parameterized robust equivariant exchange). *Let $s \geq 3$ be a prime power, let $k \geq 1$, and let $r \geq 0$. Put*

$$N(s) = \frac{s(s-1)}{2}, \qquad W(k) = \left\lfloor \frac{(k-1)^2}{4} \right\rfloor.$$

If

$$W(k) + r + 1 \leq N(s), \tag{12}$$

then deleting any selected nonfixed conjugate orbit from a ϕ-invariant $(k+2)$-arc in $\mathrm{PG}(2, s^2)$ *leaves at least r alternate-orbit repairs. The obstruction envelope $W(k)$ is exact over all profiles $k = f + 2e$. In particular, if $s \geq 4$ and $1 \leq k \leq s + 1$, every such deletion has an alternate repair.*

Proof. For a remainder profile $k = f + 2e$, the noninvariant old-secant obstruction is

$$M = fe + e(e-1) = e(k - e - 1).$$

The concave quadratic has the exact profile maximum $M \leq W(k)$: equality is attained by $f = 0$ when k is even and by $f = 1$ when k is odd. Condition (12) implies $k \leq 2s + 1$. For $s \geq 3$ this gives $k < s^2 + 1$; the completed-square occupation identity

$$2E_{\text{occ}} = (s+1)^2 + k - (f - s - 1)^2$$

then shows that at least one fixed carrier is empty. On that carrier, Theorem 3.1 gives

$$N_{\text{pair}}(D) \geq N(s) - M \geq N(s) - W(k) \geq r + 1.$$

The erased orbit is one of these legal pairs, so excluding it leaves at least r alternatives.

For the rectangular corollary, $k \leq s + 1$ gives $4W(k) \leq (k-1)^2 \leq s^2$, while $s \geq 4$ gives $s^2 + 8 \leq 2(s^2 - s) = 4N(s)$. Hence $W(k) + 2 \leq N(s)$, which is (12) with $r = 1$. The restriction $s \geq 4$ is genuine for this rectangle: $(s, k) = (3, 4)$ misses the inequality by one. □

Corollary 6.5 (All base orders from five). *Let $s \geq 5$ be a prime power. Every Frobenius-invariant eight-arc in* $\mathrm{PG}(2, s^2)$ *admits an extension by a fresh nonfixed point together with its Frobenius conjugate.*

Proof. The case $s = 5$ is Theorem 5.4. There is no prime power strictly between 5 and 7, and the final assertion of Corollary 6.1 treats every $s \geq 7$. $\square$

Proof of Theorem 1.1. The extension assertion is Corollary 6.5. Theorem 5.4 and Corollary 5.5 give (i), while Proposition 5.3 gives (ii), including the normalization scope. Theorem 6.2 and Corollary 6.3 give (iii). $\square$

The scale in (11) is the classical Lunelli–Sce scale for ordinary complete arcs, with the same secant-covering mechanism in the Frobenius quotient. Ng and Wild's line-covering problem [16] is another adjacent square-root-scale result. The point of Corollary 6.1 is the weaker pair-saturation hypothesis, not a new constant or a sharp family.

7 Formal verification

This section records the correspondence between the paper statements and their formal evidence; it is not a second proof narrative. The structural reductions are ordinary Lean proofs. A separate Mathlib-only package checks the exceptional two-fixed-point Q25 census in the normalized coordinate model used inside the proof of Proposition 5.3; the projective normalizations and semantic transport are manuscript arguments. The development uses Lean [8] on top of mathlib [18] and is divided as follows.

1. The projective-conjugation and quadratic-Frobenius modules construct the semilinear projective involution, prove the degree-two fixed-locus normalization, and count the $N = (s^2 - s)/2$ candidates on a fixed line.

2. The line-counting and forbidden-candidate modules prove Lemma 2.4, the exact secant-orbit count, the injective charge, and the semantic arc-extension conclusion in Theorem 3.1.

3. The global-count module defines the finite set of fresh nonfixed Frobenius pairs whose union with the old arc remains an arc. It proves that this semantic set is the disjoint union of the carrierwise legal sets and that its cardinality is the abstract legal count.

4. The alternate-repair module formalizes deletion of an arbitrary selected nonfixed orbit, proves that the erased orbit is one semantic legal pair of the invariant eight-arc remainder, and supplies the certificate-free eight-repair baseline. Its order-five wrapper proves Corollary 5.2.

5. The profile-envelope modules prove the five closed forms, their piecewise lower envelope, the semantic envelope comparison in Theorem 6.2, and the uniform 318-repair conclusion in Corollary 6.3.

6. The phase-diagram and parameterized-repair modules prove the exact maximum $W(k) = \lfloor (k-1)^2/4 \rfloor$, derive the empty-carrier range from the phase inequality itself, and prove Theorem 6.4 for semantic legal-pair finsets.

7. The collision and invisibility modules prove Proposition 4.1, Corollary 4.2, the fixed-center interpretation, and the cross-pair estimate (9).

8. Separate profile modules prove the lower bounds 5 and 4 for $f = 0$ and $f = 4$. For $f = 2$, residual composition certificates prove the lower bound 32 on all 1189 representatives; orbit–stabilizer, strict-composition, and exhaustive-dispatch modules prove attainment and equality-orbit exhaustion in the package's normalized coordinate model. The proof

of Proposition 5.3 carries these conclusions through both normalizations. The order-five alternate-repair module and the paper-level parity split then give Theorem 5.4 and Corollary 5.5.

9. The orbit-saturation module proves the denominator-free inequality $2s(s-1) \le (k-1)^2$ used in Corollary 6.1. The final ceiling notation and the geometric case split are paper proofs.

Corollary 3.2 is a manuscript-level specialization of the semantic carrier count to $(s, f, e) = (11, 6, 0)$. Its arithmetic and the observation $M = 0$ are printed in full; no separate Lean terminal is claimed for the number 4180.

The scoped structural builds and the sealed certificate package's declaration-level axiom audits report exactly

$$[\texttt{propext}, \texttt{Classical.choice}, \texttt{Quot.sound}],$$

the standard classical quotient profile inherited from mathlib. No public theorem depends on an admitted proof, a custom axiom, or `native_decide`. The finite leaves supporting Proposition 5.3 use kernel `decide`. The certificate terminal is the normalized classification; the passage to semantic arcs is the manuscript proof and does not rely on a count of semantic arcs.

The certificate aggregate exposes `PairCertificate.allRows`, `ResidualMinimumOrbits.card_minimumOrbitUnion`, and the two normalized exhaustion terminals in `Exhaustion.lean`.

For reproducibility, the paper-facing declarations are located as follows.

Source module	Paper-facing declaration or role
`FiniteGeom/BaerCompletion/PairExtension.lean`	`quadraticBaer_pairExtension_lowerBound`
`RelativeConicArcs/QuadraticForbidden.lean`	`exists_quadratic_pair_extension`
`RelativeConicArcs/QuadraticGlobalCount.lean`	`card_globalLegalPairs_eq_legalCount`
`RelativeConicArcs/AlternateOrbitRepair.lean`	`eight_le_alternateLegalPairs_of_seven_le`
`RelativeConicArcs/AlternateOrbitRepairProfileEnvelope.lean`	five closed forms and the piecewise profile envelope
`RelativeConicArcs/AlternateOrbitRepairProfileEnvelopeResult.lean`	`profileEnvelope_le_card_globalLegalPairs_of_card_eight`, `three_hundred_eighteen_le_alternateLegalPairs_of_seven_le`
`RelativeConicArcs/AlternateOrbitRepairPhaseDiagram.lean`	`exists_profile_attaining_worstObstruction`, `phase_profile_product_lowerBound`
`RelativeConicArcs/ParameterizedAlternateOrbitRepair.lean`	`card_alternateLegalPairs_ge_of_phase`
`RelativeConicArcs/QuadraticInvisible.lean`	aggregate equality/excess and fixed-center criteria
`FiniteGeom/BaerCompletion/OrbitSaturation.lean`	`orbitSaturation_quadratic_bound_of_split`

Artifact and reproduction

The exceptional finite layer uses a separately versioned Mathlib-only certificate package. Its exact commit and manifest digest are recorded in the verification pin. That package pins Lean 4.32.0-rc1 and mathlib revision `571b8a8e5421`. From the package root its sealed aggregate and source-reproduction checks are run by

```
nix run .#verify
```

The exact pin and trust boundary are recorded in `verification/q25-certificate-pin.json` and `verification/README.md`. The standalone paper archive does not vendor the shared human-scale Lean development listed above; its structural theorem sources and the separately pinned finite certificate have distinct release boundaries. The manuscript PDF is reproduced from its source directory with

```
make check
```

8 Conclusion

The fixed mate line of a quadratic Frobenius orbit turns symmetry-preserving arc extension into a carrierwise counting problem. The uniform count is a first-order obstruction bound; its exact correction records which secant orbits disappear at fixed centers and which visible obstructions collide. Together with one finite exceptional-profile certificate, this proves fresh conjugate-pair extension for every invariant eight-arc over every prime-power base order $s \geq 5$. In $\mathrm{PG}(2, 25)$ every such arc has at least four legal pairs, hence every selected nonfixed-orbit deletion from an invariant ten-arc has at least three alternate repairs. In the exceptional two-fixed-point profile the exact minimum is 32, with all equality cases accounted for by five residual-group orbits up to normalization.

For $s \geq 7$ the same mechanism gives at least 319 legal pairs and 318 alternate repairs. The parameterized exchange theorem identifies the general phase $\lfloor (k-1)^2/4 \rfloor + r + 1 \leq s(s-1)/2$. In coding language, these are paired two-column extensions and replacements of dimension-three MDS codes. The finite certificate closes only the exceptional Q25 profile; the carrier count and collision correction explain the general phenomenon.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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